

Neuroscientific Frontiers — Module 01: Systems —

Study Questions

1. How does the cortical hierarchy map onto the generative model? What does each level represent?
2. What signals do feedforward connections carry? What signals do feedback connections carry?
3. Which cortical layers generate prediction errors? Which generate predictions? Name the cell types.
4. What evidence supports the claim that feedforward connections are "driving" while feedback connections are "modulatory"?
5. What role do NMDA vs. AMPA receptors play in the feedforward/feedback asymmetry?
6. Describe the canonical microcircuit for predictive coding (Bastos et al., 2012). What happens in each layer?
7. How does the basal ganglia implement policy selection? What role does the striatum play?
8. What is the cerebellum's role as a forward model? How do climbing fibers signal prediction errors?
9. How does the thalamus function as a precision gate? What is the role of the reticular nucleus?
10. How does the hippocampus implement temporal inference? How does it support episodic memory?
11. What role does dopamine play in Active Inference? How does it relate to precision of reward prediction errors?
12. How does norepinephrine relate to expected precision and environmental volatility?
13. What is acetylcholine's role in sensory prediction error precision? How does this relate to attention?
14. How does serotonin relate to temporal discounting and long-horizon inference?
15. How does the neuromodulatory framework explain psychiatric symptoms as inferential failures?
16. Compare the canonical microcircuit with Rao & Ballard's (1999) original predictive coding model. What advances has Active Inference made?

17. What is the relationship between cortical column computation and a single node in a factor graph?
18. How does cortical folding (sulci and gyri) relate to the hierarchical depth of the generative model?
19. What happens to Active Inference when the cortical hierarchy is disrupted (e.g., lesion, anesthesia)?
20. Critically evaluate: Is there sufficient empirical evidence to confirm the canonical microcircuit hypothesis, or is it still largely theoretical?